The Number of King Sets of Caterpillar Digraphs with Regular Legs *

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Abstract

A king set K of a digraph is a subset of its vertices for which every vertex not in K can be reached from a vertex of K by a directed path of length at most two, and such that no two vertices of K are adjacent. A caterpillar is an undirected tree with the property that the deletion of all pendant vertices results in a path. In this paper, we give formulae for counting the number of king sets of digraphs whose underlying graphs are caterpillars with regular legs.

1 Introduction

Throughout this paper, unless otherwise mentioned, we assume that a digraph $D = (V, A)$ has no loops and no multiple arcs (including pairs of opposite arcs), and the underlying graph of D is connected, where $V = V(D)$ and $A = A(D)$ are the vertex set and the arc set of D, respectively. For an arc $(x, y) \in A$, we write $x \to y$ to mean that x dominates y (or y is dominated by x).

A vertex $x \in V(D)$ is said to be a *king* if every vertex of D can be reached from x via a path of length at most two. Maurer [4] first introduced the notion of kings on a special class of digraphs called *tournaments*, and it has been known for more than half a century that every tournament has at least one king. Due to a large amount of literatures have been devoted to the research on the topic of tournaments, many results related to kings on tournaments have been achieved (e.g., see [5] for a comprehensive survey paper). However, most of these results cannot be extended to apply in arbitrary digraphs. Recently, Bowser and Cable [2] generalized the concept of kings to that of king sets (defined below), and characterized the digraphs with a unique king set and the digraphs with a unique minimal king set. The idea of king sets is the same as *semi-kernel* that was introduced by Chvátal and Lovász in [1]. Moreover, Chvátal and Lovász also proved a result that implies that every digraph has at least one king set. In this paper, we follow the use of the term employed by Bowser and Cable [2] and focus on the problem of counting the number of king sets on caterpillar digraphs.

A *caterpillar* is an undirected tree for which the vertices that are not pendant vertices (also called *legs*) induce a path (also call a *spine*). A *caterpillar digraph* is a digraph whose underlying graph is a caterpillar. Here we use the same terminologies, such as legs and spine, for caterpillar digraphs. Let D be a caterpillar digraph. The spine of D is denoted by $sp(D)$. An *in-leg* (respec-

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tively, *out-leg*) of a vertex $v \in sp(D)$ is a leg that dominates v (respectively, is dominated by v). We use $d^-(v)$ and $d^+(v)$ to denote the number of in-legs and the number of out-legs of v, respectively. Furthermore, a specific caterpillar digraph D with $sp(D) = \{v_1, v_2, \ldots, v_n\}$ that induces the path $v_1 \to v_2 \to v_3 \to \cdots \to v_{n-1} \to v_n$ is denoted by $R_n(v_1, v_2, \ldots, v_n)$. For convenience, we sometimes say $sp(D)$ to mean its induced path. Also, for simplicity we write R_n to mean $R_n(v_1, v_2, \ldots, v_n)$ if no ambiguity arises, and for explicitness we write R_n^d (respectively, R_n^{+d}) instead of R_n if $d^-(v) + d^+(v) = d$ (respectively, $d^-(v) = 0$ and $d^+(v) = d$) for every vertex $v \in sp(D)$.

A *king set* K of a digraph $D = (V, A)$ is a subset of V satisfying the following two properties: (i) for every vertex $y \in V - K$, there exists a vertex $x \in K$ such that y can be reached from x by a directed path of length at most two; and (ii) the set $\{x \to y : x, y \in K\}$ is empty. Accordingly, Bowser and Cable call (i) the *ruling property* or *domination property* and (ii) the *royal courtesy property* (see [2] and [3]). For a digraph D, we use $k(D)$ to denote the number of distinct king sets of D.

In this paper, we restrict our attention to labelled caterpillar digraphs D with regular legs (i.e., for every vertex in the spine, the number of its legs equals to a constant, say d). Then, we give formulae to compute the number of king sets of D.

2 The number of king sets of R_n^d

A vertex x is called a *source* in a digraph D if no vertex of D dominates x. Obviously, every source in a digraph must be contained in any king set. The following general lemma provided in [3] is useful for the examination of caterpillar digraphs.

Lemma 1 [3] *If D_1 and D_2 are digraphs such that $V(D_1) \cap V(D_2) = \{x\}$, where x is a source in both D_1 and D_2, then $k(D_1 \cup D_2) = k(D_1)k(D_2)$.*

We start with a recurrence relation for the digraph R_n^{+d}. Let $c(n)$ denote the number of king sets of R_n^{+d} containing v_n.

Lemma 2 *For any digraph R_n^{+d}, $c(n) = 2^d(c(n-2) + c(n-3))$ if $n \geq 4$.*

Proof. Suppose that K is a king set of R_n^{+d} such that $v_n \in K$. Then, by the royal courtesy property, $v_{n-1} \notin K$. The absence of v_{n-1} implies, by the domination property, that K must contain either v_{n-2} or v_{n-3} and, again by the royal courtesy property, K cannot contain both of them. If $v_{n-2} \in K$, then every leg of v_{n-1} can be chosen to be a member of K or not. Thus, there are 2^d possible ways for this case. On the other hand, if $v_{n-3} \in K$, then all legs of v_{n-1} must be contained in K. Further, there are 2^d possible ways that we can choose from d legs of v_{n-2} to become the members of K in this case. Hence, the equation follows. □

Obviously, $c(1) = 1$ and $c(2) = 0$. If we set $c(0) = 0$, the recurrence relation as stated in Lemma 2 can be extended to hold for $n \geq 3$. As to the recurrence relation, we are easy to check that the characteristic equation $r^3 - 2^d r - 2^d = 0$ has the following roots:

$$\begin{aligned} r_1 &= \alpha + \beta, \\ r_2 &= -\frac{(\alpha+\beta)}{2} - \frac{\sqrt{3}(\alpha-\beta)}{2} i, \\ r_3 &= \overline{r_2}. \end{aligned}$$

where

$$\alpha = \frac{2^{(d-1)/3}}{\sqrt{3}} \left(\sqrt{27} - \sqrt{27 - 2^{d+2}}\right)^{1/3}$$

and

$$\beta = \frac{2^{(d-1)/3}}{\sqrt{3}} \left(\sqrt{27} + \sqrt{27 - 2^{d+2}}\right)^{1/3}.$$

So, we have $\alpha\beta = 2^d/3$.

The solution of the recurrence relation is $c(n) = x_1 r_1^n + x_2 r_2^n + x_3 r_3^n$, where x_1, x_2, and x_3 are constants determined by the initial conditions $c(0) = c(2) = 0$ and $c(1) = 1$. Substitute $n = 0, 1, 2$ to obtain a 3×3 system of simultaneous equations. Then, solving these equations for x_1, x_2, and x_3, we get

$$\begin{aligned} x_1 &= \frac{r_1^2}{2^d(2r_1+3)}, \\ x_2 = \overline{x_3} &= \frac{1}{2(2r_1+3)} \left(\frac{-r_1^2}{2^d} + \frac{r_1+3}{\sqrt{3}(\alpha-\beta)} i\right). \end{aligned}$$

Theorem 1 *For all integers $n \geq 1$, $k(R_n^{+d}) = c(n+3)/2^d$.*

Proof. Suppose that K is a king set of R_n^{+d} for $n \geq 3$ and let $x \in K$ be the vertex with the largest index in the spine. By the domination property, $x \in \{v_n, v_{n-1}, v_{n-2}\}$. If $x = v_{n-1}$, there are 2^d possible ways for choosing the legs of v_n to be the members of K. If $x = v_{n-2}$, all legs of v_n must belong to K, and again there are 2^d possible ways for choosing the legs of v_{n-1} to be the members of K. Thus $k(R_n^{+d}) = c(n) + c(n-1)2^d + c(n-2)2^d$. Now, we can apply the recurrence of Lemma 2 twice to produce the desired equation. $\square$

We now deal with the number of king sets of R_n^d and suppose that there exists some vertex $v_i \in sp(R_n^d)$ with $d^-(v_i) \neq 0$. Let v_p be the vertex with the smallest index in $sp(R_n^d)$ for which every vertex v_j with $j > p$ has no in-legs. Since there exist some in-legs connecting R_n^d at v_p, if K is any king set of R_n^d then such in-legs must belong to K. Furthermore, by the royal courtesy property, $v_p \notin K$. In this case, we can decompose R_n^d at v_p into two sub-digraphs D_1 and D_2 as follows:

$$D_1 \equiv R_{p-1}^d(v_1, v_2, \ldots, v_{p-1}) \text{ and}$$
$$D_2 \equiv R_{n-p+1}^d(v_p, v_{p+1}, \ldots, v_n) \qquad (1)$$

Let D_2' be the digraph obtained from D_2 by contracting the in-legs of v_p to form a new vertex, say v_c. Then, we can view D_2' as a caterpillar digraph $R_{n-p+2}(v_c, v_p, v_{p+1}, \ldots, v_n)$ where v_c is the source in the spine of D_2'. For example, Figure 1(a) shows a caterpillar digraph R_6^3 decomposed at v_4 where v_a and v_b are in-legs. Figure 1(b) and 1(c) show the two corresponding sub-digraphs D_1 and D_2', respectively. In D_2', v_c comes from the contraction of v_a and v_b.

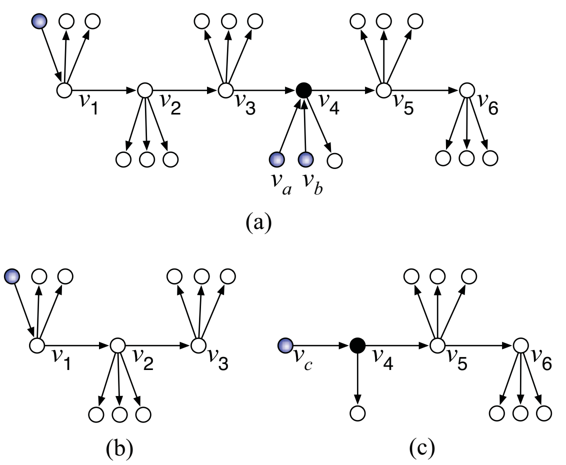

Figure 1: (a) A decomposition of caterpillar digraph R_6^3 at v_4 which contained two in-legs; (b) Sub-digraph D_1; (c) Sub-digraph D_2'.

Lemma 3 *For $n \geq p \geq 1$, $k(D_2') = 2^{d^+(v_p)-d}k(R_{n-p+2}^{+d})$.*

Proof. Let v_1^* and v_2^* be the first two vertices in $sp(R_{n-p+2}^{+d})$. Since $sp(D_2')$ and $sp(R_{n-p+2}^{+d})$ have the same length, the differences between D_2' and R_{n-p+2}^{+d} are the following: (i) $d^+(v_c) = 0 \neq d^+(v_1^*) = d$; and (ii) $d^+(v_p) < d^+(v_2^*) = d$. Since v_1^* is a source in R_{n-p+2}^{+d}, the legs of v_1^* cannot belong to any king set. Thus, difference (i) does not cause any effect for computing the number of king sets. We consider difference (ii) as follows. Obviously, there exist $2^{d^+(v_p)}$ possible ways for choosing the legs of v_p to be the members of a king set in D_2', and contrastively, there exist 2^d possible ways for choosing the legs of v_2^* to be the members of a king set in R_{n-p+2}^{+d}. Therefore, the proportion of $k(D_2')$ and $k(R_{n-p+2}^{+d})$ follows from the ratio. $\square$

To compute $k(R_n^d)$, we first consider the extreme case that only the vertex $v_1 \in sp(R_n^d)$ has in-legs. In this case, D_1 is a null graph and, by definition, the empty set is the unique king set of D_1. Thus, $k(D_1) = 1$. As a consequence, $k(R_n^d) = k(D_2')$ and it can be computed by Lemma 3. In general, the following theorem shows that $k(R_n^d) = k(D_1)k(D_2')$ even if there exist more than one vertex of $sp(R_n^d)$ that have in-legs. We note that $k(D_1)$ can be determined if the decomposition processes for D_1 are carried out repeatedly until either the extreme case has been arrived or the remaining digraph of D_1 has no in-legs, where the latter case can be computed by Theorem 1.

Theorem 2 *Suppose that R_n^d is decomposed into two sub-digraphs D_1 and D_2'. Then, $k(R_n^d) = k(D_1)k(D_2')$.*

Proof. Let K be any king set of R_n^d and K' be any king set of D_2'. Suppose that the decomposition is occurred at a vertex $v_p \in sp(R_n^d)$. We know that every in-leg of v_p must belong to K, and by the royal courtesy property, $v_p \notin K$. On the other hand, since v_c is a source of D_2', we have $v_c \in K'$ and $v_p \notin K'$. Excepting v_p and its in-legs, it is obvious that the number of ways for determining the remaining vertices of R_n^d to belong to a king set or not is equal to $k(D_1)k(D_2')$. $\square$

3 The number of king sets of $R_{n,m}^d$

In what follows, we consider a more general case that the spine of a caterpillar digraph has more than one directed path. For $n, m \geq 2$, we denote $R_{n,m}(v_1, v_2, \ldots, v_{n+m-1})$ (or $R_{n,m}$ if no ambiguity arises) as a caterpillar digraph with $sp(R_{n,m}) = \{v_1, v_2, \ldots, v_{n+m-1}\}$ that induce two directed paths $v_1 \to v_2 \to \cdots \to v_{n-1} \to v_n$ and $v_{n+m-1} \to v_{n+m-2} \to \cdots \to v_{n+1} \to v_n$. Also, we denote $R^d_{n,m}$ (respectively, $R^{+d}_{n,m}$) if $d^-(v) + d^+(v) = d$ (respectively, $d^-(v) = 0$ and $d^+(v) = d$) for every vertex v in the spine. It is easy to see that the caterpillar digraph $R^d_{n,m}$ contains two sub-digraphs $R^d_n(v_1, v_2, \ldots, v_n)$ and $R^d_m(v_{n+m-1}, v_{n+m-2}, \ldots, v_n)$, where the second sub-digraph is isomorphic to $R^d_m(u_1, u_2, \ldots, u_m)$ under the mapping $u_i = v_{n+m-i}$ for $i = 1, 2, \ldots, m$.

Theorem 3 *For all integers $n, m \geq 2$,*

$$k(R^{+d}_{n,m}) = k(R^{+d}_n)c(m) + k(R^{+d}_m)c(n) - \sum_{i=0}^{2}(-1)^i c(n-i)c(m-i)2^{di}.$$

Proof. As stated in the previous, the digraph $R^{+d}_{n,m}$ contains sub-digraphs isomorphic to R^{+d}_n and R^{+d}_m. Then, all the king sets of $R^{+d}_{n,m}$ can be partitioned according to how the domination property is satisfied for v_n. Assume that K is any king set of $R^{+d}_{n,m}$. We consider the following cases.

Case 1: $v_n \in K$. In this case, $K \cap V(R^{+d}_n)$ and $K \cap V(R^{+d}_m)$ are king sets of R^{+d}_n and R^{+d}_m, respectively, containing v_n as the last vertex in its spine. Thus, it contributes $c(n)c(m)$ different kinds of king sets.

The remaining cases are considered for $v_n \notin K$.

Case 2: either $v_{n-2} \in K$ or $v_{n-1} \in K$, and $v_{n+1}, v_{n+2} \notin K$. In this case, $v_{n+3} \in K$ by the domination property. Thus, $K \cap V(R^{+d}_n)$ is one of the $2^d(c(n-1) + c(n-2))$ king sets of R^{+d}_n containing either v_{n-2} or v_{n-1} as the last vertex in its spine. On the other hand, $K \cap V(R^{+d}_m)$ is one of the $2^d c(m-3)$ king sets of R^{+d}_m containing v_{n+3} as the last vertex in its spine. Consequently, there are $2^{2d}(c(n-1)+c(n-2))c(m-3)$ such king sets.

Case 3: either $v_{n+2} \in K$ or $v_{n+1} \in K$, and $v_{n-1}, v_{n-2} \notin K$. This case is the same as the preceding case by reversing n and m. Thus, there are $2^{2d}(c(m-1) + c(m-2))c(n-3)$ additional king sets.

Case 4: $|K \cap \{v_{n-2}, v_{n-1}\}| = |K \cap \{v_{n+1}, v_{n+2}\}| = 1$. There are four conditions that we need to consider.

Case 4.1: If $v_{n-1}, v_{n+1} \in K$, then we have $2^d c(n-1)c(m-1)$ king sets.

Case 4.2: If $v_{n-1}, v_{n+2} \in K$, then we have $2^{2d}c(n-1)c(m-2)$ king sets.

Case 4.3: If $v_{n-2}, v_{n+1} \in K$, then we have $2^{2d}c(n-2)c(m-1)$ king sets.

Case 4.4: If $v_{n-2}, v_{n+2} \in K$, then we have $2^{2d}c(n-2)c(m-2)$ king sets.

Aggregating the contributions over all mutually exclusive and exhaustive cases, we have

$$\begin{aligned} k(R^{+d}_{n,m}) = \; & c(n)c(m) + \\ & 2^{2d}c(n-1)c(m-3) + \quad (2)\\ & 2^{2d}c(n-2)c(m-3) + \quad (3)\\ & 2^{2d}c(n-3)c(m-1) + \quad (4)\\ & 2^{2d}c(n-3)c(m-2) + \quad (5)\\ & 2^{d}c(n-1)c(m-1) + \\ & 2^{2d}c(n-1)c(m-2) + \quad (6)\\ & 2^{2d}c(n-2)c(m-1) + \quad (7)\\ & 2^{2d}c(n-2)c(m-2). \quad (8) \end{aligned}$$

Using Lemma 2, we combine terms (2) and (6), and terms (3) and (8) twice to obtain $c(n+1)c(m)$. Similarly, combining terms (4) and (7), and terms (5) and (8) twice, we obtain $c(n)c(m+1)$. Thus,

$$\begin{aligned} k(R^{+d}_{n,m}) &= c(n)c(m) + \\ &\quad c(n+1)c(m) + c(n)c(m+1) + \\ &\quad 2^d c(n-1)c(m-1) \\ &\quad -2^{2d}c(n-2)c(m-2) \\ &= k(R^{+d}_n)c(m) + k(R^{+d}_m)c(n) \\ &\quad -c(n)c(m) + 2^d c(n-1)c(m-1) \\ &\quad -2^{2d}c(n-2)c(m-2). \end{aligned}$$

The last equality holds since $k(R^{+d}_n) = c(n+3)/2^d = c(n) + c(n+1)$ and $k(R^{+d}_m) = c(m+3)/2^d = c(m) + c(m+1)$ follow from Theorem 1 and Lemma 2. □

We are now in a position to compute the number of king sets for $R^d_{n,m}$. Suppose that there exists some vertex $v_i \in sp(R^d_{n,m})$ with $d^-(v_i) \neq 0$. As we know that $R^d_{n,m}$ contains two sub-digraphs isomorphic to $R^d_n(v_1, v_2, \ldots, v_n)$ and $R^d_m(u_1, u_2, \ldots, u_m)$. Let v_p (respectively, u_q) be the vertex with the smallest index in $sp(R^d_n)$ (respectively, $sp(R^d_m)$) for which every vertex v_i with $p < i$ (respectively, u_j with $q < j$) has no in-legs if such an i (respectively, j) exists. Recall that we have the mapping $v_{n+m-q} = u_q$ between $R^d_{n,m}$ and R^d_m. Thus, similar to (1) we can decompose

$R^d_{n,m}$ at v_p and v_{n+m-q} into three disjoint caterpillar digraphs such that they are isomorphic to the following:

$$D_n \equiv R^d_{p-1}(v_1, v_2, \ldots, v_{p-1}) \text{ and}$$
$$D_m \equiv R^d_{q-1}(u_1, u_2, \ldots, u_{q-1})$$

and

$$D_2 \equiv R^d_{n-p+1,m-q+1}(v_p, v_{p+1}, \ldots, v_n, \ldots, v_{n+m-q}). \tag{9}$$

By contracting the in-legs of v_p and v_{n+m-q} in (9) to form the new vertices v_c and v_b respectively, we can restate the digraph as

$$D'_2 \equiv R_{n-p+2,m-q+2}(v_c, v_p, v_{p+1}, \ldots, v_n, \ldots, v_{n+m-q}, v_b).$$

It should be noted that, depending on the absence of v_p or u_q (but not the both), D_n or D_m might be a null graph. Using the method mentioned in the previous section, we can compute $k(D_n)$ and $k(D_m)$ respectively. Let $D_1 \equiv D_n \cup D_m$. Then, it is obvious that the number of king sets of D_1 is $k(D_1) = k(D_n)k(D_m)$. Since D'_2 is a caterpillar digraph with no in-legs, similar to Lemma 3 we can obtain the number of king sets of D'_2 as follows

$$k(D'_2) = 2^{(d^+(v_p)+d^+(v_q)-2d)} k(R^{+d}_{n-p+2,m-q+2})$$

where $k(R^{+d}_{n-p+2,m-q+2})$ can be directly computed by Theorem 3.

In spite of the decomposition occurred at any vertex, a proof similar to Theorem 2 can show the following.

Theorem 4 *Suppose that $R^d_{n,m}$ is decomposed into two sub-digraphs D_1 and D'_2. Then, $k(R^d_{n,m}) = k(D_1)k(D'_2)$.*

4 Concluding remark

As a consequence, Theorems 2 and 4 used with Lemma 1 provide the number of king sets for arbitrary caterpillar digraphs with regular legs.

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